

PRACTICAL WEEK 7

TASK 1: Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
jhansi_priya@Jhansi-Priya: ~/ × + v
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;      // Global variable
static int static_var = 200; // Static variable

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 300;      // Stack variable
    int *heap_var;            // Pointer on stack

    heap_var = malloc(sizeof(int));
    *heap_var = 400;

    printf("Process ID (PID)      : %d\n", getpid());

    printf("Address of code       : %p\n", (void *)code_function);
    printf("Address of global    : %p\n", (void *)&global_var);
    printf("Address of static     : %p\n", (void *)&static_var);
    printf("Address of heap       : %p\n", (void *)heap_var);
    printf("Address of stack      : %p\n", (void *)&stack_var);

    printf("\nProgram is running...\n");
    printf("Press Ctrl+C to terminate.\n");

    while (1)
    {
        sleep(1);
    }

    free(heap_var);

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya: ~/ × + v
jhansi_priya@Jhansi-Priya:~$ mkdir Practical7
jhansi_priya@Jhansi-Priya:~$ cd Practical7
jhansi_priya@Jhansi-Priya:~/Practical7$ nano task1.c
jhansi_priya@Jhansi-Priya:~/Practical7$ gcc task1.c -o task1
jhansi_priya@Jhansi-Priya:~/Practical7$ ls
task1 task1.c
jhansi_priya@Jhansi-Priya:~/Practical7$ ./task1
Process ID (PID)      : 530
Address of code       : 0x560f3248f1c9
Address of global     : 0x560f32492010
Address of static     : 0x560f32492014
Address of heap       : 0x560f57d292a0
Address of stack      : 0x7ffea91c930c

Program is running...
Press Ctrl+C to terminate.
|
```

```
jhansi_priya@Jhansi-Priya:~/Practical7$ cat /proc/623/maps
60b270588000-60b270589000 r--p 00000000 08:30 42119
/home/jhansi_priya/Practical7/task1
60b270589000-60b27058a000 r-xp 00001000 08:30 42119
/home/jhansi_priya/Practical7/task1
60b27058a000-60b27058b000 r--p 00002000 08:30 42119
/home/jhansi_priya/Practical7/task1
60b27058b000-60b27058c000 r--p 00002000 08:30 42119
/home/jhansi_priya/Practical7/task1
60b27058c000-60b27058d000 rw-p 00003000 08:30 42119
/home/jhansi_priya/Practical7/task1
60b27609f000-60b2760c0000 rw-p 00000000 00:00 0
[heap]
7019c1a00000-7019c1a28000 r--p 00000000 08:30 49765
/usr/lib/x86_64-linux-gnu/libc.so.6
7019c1a28000-7019c1bb0000 r-xp 00028000 08:30 49765
/usr/lib/x86_64-linux-gnu/libc.so.6
7019c1bb0000-7019c1bff000 r--p 001b0000 08:30 49765
/usr/lib/x86_64-linux-gnu/libc.so.6
7019c1bff000-7019c1c03000 r--p 001fe000 08:30 49765
/usr/lib/x86_64-linux-gnu/libc.so.6
7019c1c03000-7019c1c05000 rw-p 00202000 08:30 49765
/usr/lib/x86_64-linux-gnu/libc.so.6
7019c1c05000-7019c1c12000 rw-p 00000000 00:00 0
7019c1cc2000-7019c1cc5000 rw-p 00000000 00:00 0
7019c1ccc000-7019c1cce000 rw-p 00000000 00:00 0
7019c1cce000-7019c1cd2000 r--p 00000000 00:00 0
```

TASK2: Make use of the /proc/<PID>/maps file and memory analysis tools to study the memory organization of a running process.

```
jhansi_priya@Jhansi-Priya: ~/ × + ▾ □
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
static int static_var = 200;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 300;
    int *heap_var;

    heap_var = malloc(sizeof(int));
    *heap_var = 400;

    printf("Process ID (PID)      : %d\n", getpid());

    printf("Address of code      : %p\n", (void *)code_function);
    printf("Address of global     : %p\n", (void *)&global_var);
    printf("Address of static     : %p\n", (void *)&static_var);
    printf("Address of heap       : %p\n", (void *)heap_var);
    printf("Address of stack      : %p\n", (void *)&stack_var);

    while (1)
    {
        sleep(1);
    }

    free(heap_var);

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya:~/Practical7$ nano task2.c
jhansi_priya@Jhansi-Priya:~/Practical7$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/Practical7$ ./task2
Process ID (PID)      : 618
Address of code       : 0x6369c48a81c9
Address of global     : 0x6369c48ab010
Address of static     : 0x6369c48ab014
Address of heap       : 0x6369c4e5f2a0
Address of stack      : 0x7ffd3c052fec
^C
```


